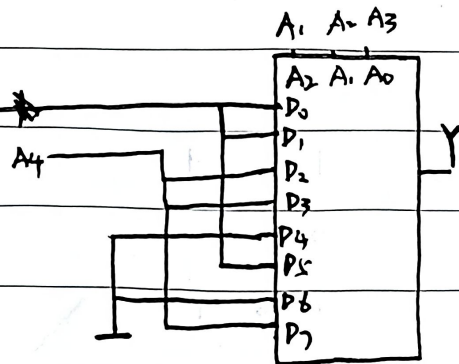


# 春四周 作业4

1. A: 01  
B: 10  
AB: 11  
O: 00

$A_1 A_2$  输入血型  $A_3 A_4$  受血者血型  $Y=1$  符合规定  $Y=0$  不符合

|                     | $A_1 A_2$ | $A_3 A_4$ | $Y$ |                                                                                     |
|---------------------|-----------|-----------|-----|-------------------------------------------------------------------------------------|
| $A \rightarrow A$   | 0 1       | 0 1       | 1   | $Y = \bar{A}_1 \bar{A}_2 + \bar{A}_1 A_2 A_4 + A_1 \bar{A}_2 A_3 + A_1 A_2 A_3 A_4$ |
|                     | 0 1       | 1 0       | 0   |                                                                                     |
| $A \rightarrow AB$  | 0 1       | 1 1       | 1   | $A_1 A_2 A_3$ 作数据地址                                                                 |
|                     | 0 1       | 0 0       | 0   |                                                                                     |
| $B \rightarrow B$   | 1 0       | 0 1       | 0   |                                                                                     |
| $B \rightarrow AB$  | 1 0       | 1 0       | 1   |                                                                                     |
|                     | 1 0       | 1 1       | 1   |                                                                                     |
|                     | 1 0       | 0 0       | 0   |                                                                                     |
|                     | 1 1       | 0 1       | 0   |                                                                                     |
|                     | 1 1       | 1 0       | 0   |                                                                                     |
| $AB \rightarrow AB$ | 1 1       | 1 1       | 1   |                                                                                     |
|                     | 1 1       | 0 0       | 0   |                                                                                     |
| 全符合                 | 0 0       | 0 1       | 1   |                                                                                     |
|                     | 0 0       | 1 0       | 1   |                                                                                     |
|                     | 0 0       | 1 1       | 1   |                                                                                     |
|                     | 0 0       | 0 0       | 1   |                                                                                     |



2. MODE A B C X Y Z

(1)

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 |

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 |

(2)

MODE A | B C 00 01 11 10

|    |    |   |   |   |   |
|----|----|---|---|---|---|
| X: | 00 | 0 | 0 | 1 | 0 |
|    | 01 | 1 | 0 | X | X |
|    | 11 | 0 | 1 | X | X |
|    | 10 | 0 | 0 | 0 | 0 |

$$X = \overline{\text{MODE}} \overline{A} \overline{C} + \overline{\text{MODE}} B C + \text{MODE} A C + \overline{\text{MODE}} \overline{A} \overline{B} \overline{C}$$

Y:

| MODE A \ BC | 00 | 01 | 11 | 10 |
|-------------|----|----|----|----|
| 00          | 0  | 1  | 0  | 1  |
| 01          | 0  | 0  | X  | X  |
| 11          | 1  | 0  | X  | X  |
| 10          | 0  | 0  | 1  | 0  |

$$Y = \overline{MODE} \bar{A} \bar{B} C + MODE BC + \overline{MODE} B \bar{C} + MODE A \bar{C}$$

z:

| MODE A \ BC | 00 | 01 | 11 | 10 |
|-------------|----|----|----|----|
| 00          | 1  |    |    | 1  |
| 01          | 1  |    | X  | X  |
| 11          | 1  |    | X  | X  |
| 10          | 1  |    |    | 1  |

$$z = \bar{C}$$

00